Atomic structure

Model the atom

1 Particles & isotopes

Tiny positive nucleus: protons + neutrons; negative electrons in energy levels.

Z = protons; A = protons + neutrons. Isotopes differ in neutrons. Losing electrons forms positive ions.

2 Evidence changes models

Electron → plum pudding. Alpha scattering → tiny dense nucleus, mostly empty space.

Bohr: specific levels; Chadwick: neutrons. Absorb radiation → electron rises; emit → falls.

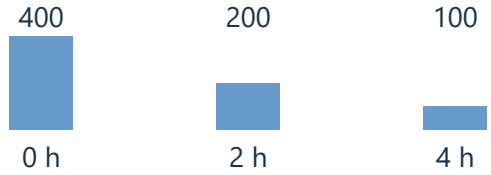
Track nuclear changes

3 Decay & equations

Random nuclear decay. Activity in **Bq = decays/s;** detector count rate may be lower.

Alpha: A−4, Z−2. Beta-minus: A unchanged, Z+1. Gamma: A and Z unchanged.

4 Half-life



Count rate halves every 2 h

Time for activity/undecayed nuclei to halve. After n half-lives, **$(½)^n$ remains.**

Subtract background first. After 3 half-lives, 1/8 remains and 7/8 has decayed.

Use radiation wisely

5 Hazards & uses

Contamination = radioactive material present; irradiation = exposure. Alpha especially harmful inside.

Gamma tracers; targeted treatment. Match half-life and penetration to use; balance benefit and dose.

6 Fission & fusion

Fission: large nucleus absorbs neutron → splits, releasing energy, neutrons and gamma.

Neutrons can cause a chain reaction. Fusion: light nuclei join; some mass becomes radiation energy.